Apply filters to SQL queries

Project description

In this project, I used SQL to investigate potential security issues within an organization's systems. I analyzed data from the employees and log_in_attempts tables, applying filters with AND, OR, and NOT operators to identify suspicious login attempts, unusual access patterns, and potential device-related risks. Through these queries, I was able to isolate relevant records and support security monitoring efforts, demonstrating my ability to use SQL for practical cybersecurity analysis and threat detection.

Retrieve after hours failed login attempts

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00' AND success = 0;
```

This query selects all records from the log_in_attempts table where the login occurred after 18:00 and the attempt failed. It uses the AND operator to combine both conditions, ensuring only unsuccessful logins outside business hours are returned.

Retrieve login attempts on specific dates

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-08' OR login_date = '2022-05-09';
```

This query retrieves all login attempts that occurred on either 2022-05-08 or 2022-05-09. The OR operator is used to include records from both dates, allowing investigation of activity around the suspicious event.

Retrieve login attempts outside of Mexico

```
SELECT *
FROM log_in_attempts
WHERE NOT country LIKE 'MEX%';
```

This query returns all login attempts that did not originate from Mexico. It uses NOT with the LIKE 'MEX%' pattern to exclude any country values starting with "MEX," covering both "MEX" and "MEXICO."

Retrieve employees in Marketing

```
SELECT *
FROM employees
WHERE department = 'Marketing' AND office LIKE 'East%';
```

This query selects all employees in the Marketing department who are located in offices within the East building. It combines conditions using AND and uses LIKE 'East%' to match all East office locations.

Retrieve employees in Finance or Sales

```
SELECT *
FROM employees
WHERE department = 'Finance' OR department = 'Sales';
```

This query retrieves all employees who belong to either the Finance or Sales departments. The OR operator allows inclusion of both departments in a single query result.

Retrieve all employees not in IT

```
SELECT *
FROM employees
WHERE NOT department = 'Information Technology';
```

This query selects all employees who are not part of the Information Technology department. It uses the NOT operator to exclude IT employees, returning only those who still require the update.

Summary

This project focused on using SQL queries to investigate a potential security incident and support system updates. I analyzed login records to identify suspicious patterns, such as failed after-hours access and activity on specific dates, and filtered out irrelevant data to narrow the investigation. I also queried employee data to identify specific groups requiring security updates based on department and location. Overall, the project highlights practical use of SQL for both incident investigation and operational security tasks.